ANALYSIS OF E-COMMERCE DATA

DATA COLLECTION & DATA CLEANING PROCESS STEPS

- Data successfully collected from the given source.
- Created a **Table** for each data set in **Excel**. (Customer Data, Product Data, Store Data & Sales Data.)
- By using **Excel** all the data from each table were cleaned by removing the **Duplicate** values, corrected the **Formatting Errors** with the help of **Number Format** option and replaced few data by using **Replace Value** option.
- Changed the data type for **Date** and **Currency**, **Text** and **Number** columns and **Imputed** all the missing data in each cell for all the table using **Excel** and also all the imputed cells were **Highlighted**.
- Imputed the data for loyalty level column from **customer_db** table.
- Imputed values for **Cost** column and **Stock** column in **product_db** table.
- Imputed values for **Quality** column, **Unit price** column, Discount column and Total amount column in **sales_db** table.
- Created a new columns in sales_db table as **Sales Performance**.
- Created and new columns as **Category**, **Product Name**, **Region**, **Loyalty Level** & **Month** and combined these columns in **Sales_db** table using **VLOOKUP** function.

DESCRIPTIVE STATISTICS

Created a **Descriptive Statistics** for Total Amount Final column from the table sales_db.

DESCRIPTIVE STATISTICS	
Total Amount Final	
Mean	1086.509174
Standard Error	18.29755471
Median	870.1885
Mode	#N/A
Standard Deviation	818.2915231
Sample Variance	669601.0168
Kurtosis	0.061186226
Skewness	0.882150441
Range	3889.7838
Minimum	18.459
Maximum	3908.2428
Sum	2173018.348
Count	2000

PIVOT TABLE

Created a **Pivot Table** in excel as category wise total revenue, loyalty level, month wise trends, region wise sales performance, payment type & top products.

CATEGORY WISE TOTAL REVENUE

Row Labels	Sum of Total amount Final
Beauty	$3,22,028.91
Clothing	$3,78,484.12
Electronics	$5,05,240.18
Home	$4,25,254.97
Sports	$5,42,010.17
Grand Total	**$21,73,018.35**

REGION WISE SALES PERFORMANCE

Row Labels	Count of Sales Performance
East	590
North	596
South	581
West	233
Grand Total	**2000**

PAYMENT TYPE

Row Labels	Count of Sales_ID
COD	660
Credit Card	645
PayPal	695
Grand Total	**2000**

LOYALTY LEVEL

Row Labels	Sum of Total amount Final
Bronze	$5,20,057.95
Gold	$5,18,117.42
Platinum	$6,72,701.01
Silver	$4,44,629.63
Uncategorized	$17,512.34
Grand Total	**$21,73,018.35**

MONTH WISE TRENDS

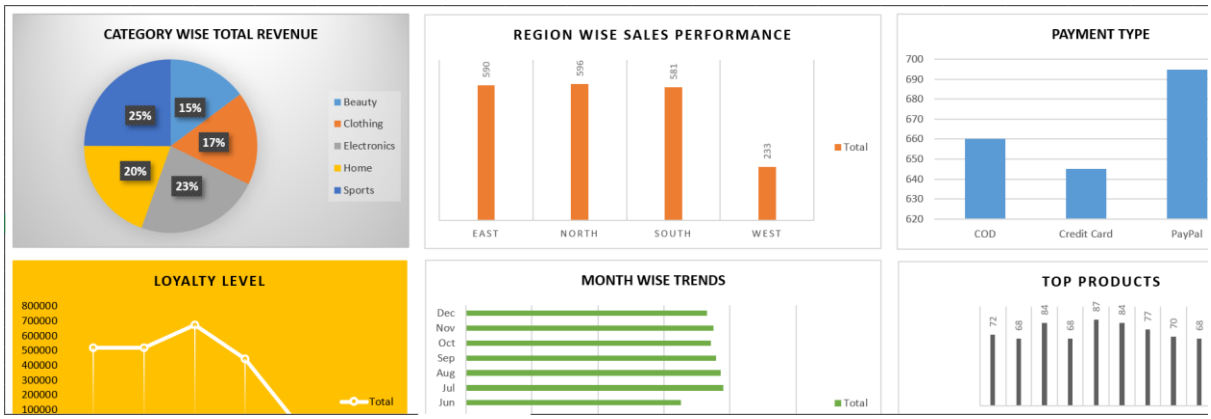
Row Labels	Sum of Total amount Final
Jan	$1,73,199.85
Feb	$1,55,073.94
Mar	$2,10,453.79
Apr	$1,92,881.46
May	$1,46,199.31
Jun	$1,62,640.02
Jul	$1,94,711.82
Aug	$1,92,878.04
Sep	$1,89,613.42

TOP PRODUCTS

Row Labels	Sum of Quantity Final
4K Home Projector	72
Face Foundation Kit	68
Gaming Mechanical Keyboard	84
Laptop Cooling Pad	68
Men's Casual Sneakers	87
Portable Power Bank	84
Shin Guards – Adult	77
Soccer Training Cone Set	70
Vitamin E Body Lotion	68

DATA VISUALIZATION

Created a Data Visualization in excel for all the pivot tables.



DASHBOARD

FUNCTIONS USED FOR DATA IMPUTATION

- =IF(ISBLANK([@[Loyalty_Level]]),"Uncategorized",[@[Loyalty_Level]])
- =IF(ISBLANK([@Cost]),AVERAGEIFS([Cost],[Category],[@Category],[Sub_Category],[@[Sub_Category]]),[@Cost])
- =IF(ISBLANK([@Stock]),AVERAGEIFS([Stock],[Category],[@Category],[Sub_Category],[@[Sub_Category]]),[@Stock])
- =IF(ISBLANK([@Quantity]),AVERAGEIF([Product_ID],[@[Product_ID]],[Quantity]),[@Quantity])
- =IF(ISBLANK([@[Unit_Price]]),AVERAGEIF([Product_ID],[@[Product_ID]],[Unit_Price]),[@[Unit_Price]])
- =IF(ISBLANK([@Discount]),AVERAGEIF([Payment_Type],[@[Payment_Type]],[Discount]),[@Discount])
- =[@[Unit price Final]]*[@[Quantity Final]]*(1-[@[Discount Final]])
- =IFS([@[Total Amount Final]]>2500,"HIGH",[@[Total Amount Final]]>=1000,"MEDIUM",[@[Total Amount Final]]<1000,"LOW")
- =TEXT([@[Order_Date]],"mmm")

FUNCTIONS USED IN VLOOKUP

- =VLOOKUP(D:D,product_db,3,FALSE)
- =VLOOKUP(D:D,product_db,2,FALSE)
- =VLOOKUP(E:E,store_db,3,FALSE)
- =VLOOKUP(C:C,customer_db,10,TRUE)

Discount Fir	Payment_Type	Total_Amount	Total Amount Fin	Sales Performanc	Category	Product Name	Region	Loyalty Level Fir	Montl
10.00%	PayPal	$598.24	$598.24	LOW	Beauty	Organic Sunscreen Lotion	West	Uncategorized	May
7.00%	Credit Card	$751.59	$751.59	LOW	Home	Premium Blanket Throw	East	Uncategorized	Jun
26.00%	Credit Card	$1,087.84	$1,087.84	MEDIUM	Electronics	Noise-Canceling Headphones	South	Uncategorized	Oct
3.00%	COD	$837.24	$837.24	LOW	Electronics	HDMI Cable – 2m	South	Platinum	Jun
11.00%	PayPal	$268.10	$268.10	LOW	Electronics	Car Dashboard Phone Holder	South	Platinum	Aug
10.00%	COD	$835.85	$835.85	LOW	Electronics	Smart LED Light Strip	East	Platinum	Mar
28.00%	Credit Card	$602.25	$602.25	LOW	Home	Decorative Lamp Shade	North	Platinum	Oct
11.00%	COD	$616.55	$616.55	LOW	Sports	Sports Stopwatch Pro	North	Platinum	Aug
9.00%	Credit Card	$392.84	$392.84	LOW	Home	Kitchen Cutting Board	South	Platinum	Mar
23.00%	PayPal	$466.78	$466.78	LOW	Clothing	Women's Ankle Boots	North	Platinum	Jul
1.00%	PayPal	$445.10	$445.10	LOW	Electronics	Brother Laser Printer	South	Bronze	Aug
28.00%	Credit Card	$105.47	$105.47	LOW	Beauty	Luxury Perfume Spray	East	Bronze	Nov
9.00%	COD	$1,311.42	$1,311.42	MEDIUM	Sports	Cricket Bat	North	Bronze	Feb
2.00%	PayPal	$428.90	$428.90	LOW	Beauty	Hair Strengthening Shampoo	South	Bronze	Jul
21.00%	Credit Card	$39.97	$39.97	LOW	Electronics	Bluetooth Car Audio System	North	Bronze	Feb
13.00%	PayPal	$476.92	$476.92	LOW	Beauty	Makeup Brush Set	North	Silver	Jul
5.00%	PayPal	$680.88	$680.88	LOW	Sports	Badminton Racket Set	North	Silver	Nov
4.00%	COD	$435.27	$435.27	LOW	Beauty	Matte Lipstick Set	West	Silver	May
19.00%	PayPal	$287.39	$287.39	LOW	Home	Microwave Oven	East	Silver	Nov
17.00%	COD	$1,078.82	$1,078.82	MEDIUM	Clothing	Women's Formal Blazer	South	Silver	Jun
14.00%	Credit Card	$2,871.27	$2,871.27	HIGH	Sports	Shin Guards – Adult	South	Silver	Jul

The above colums in **Yellow** colour columns are additional combined columns using **VLOOKUP** function except the **Month** column.

KEY BUSINESS INSIGHTS

1. CATEGORY WISE SALES PERFORMANCE

The **Sports** category contributes the highest share of total sales which is

$5,42,010.17.
The **Beauty** category contributes the lowest share of total sales which is **$3,22,028.91**.

2. **REGION WISE SALES**
   The **North** region has the highest sales record which is **$6,44,042.62**.
   The **West** region has the lowest sales record which is **$2,70,607.42**.

3. **LOYALTY LEVEL WITH PAYMENT METHOD**
   The **Platinum** level customers are having the highest transaction which is **$6,72,701.01** and the most used payment type for platinum level is **PayPal** among customers.
   The **Uncategorized** loyalty level has the lowest transaction which is **$17,512.34** and the most less used payment type for uncategorized is **Credit Card**.

4. **MONTHLY SALES DISTRIBUTION**
   The **March** month has the highest contribution in total revenue which is $**2,10,453.79** and the highly purchased product in march month is **Men's casual sneakers**.
   The **May** month has the lowest contribution in total revenue which is **$1,46,199.31** and the lowest purchased products in may month are **Face foundation kit**, **Laptop cooling pad** & **Vitamin E body lotion**.

E-Commerce Data Analysis – Summary

This project focuses on analyzing an E-commerce sales dataset using Microsoft Excel. The main objective was to clean the data, analyze sales performance, and visualize important business insights through charts and dashboards.

1. Data Collection and Cleaning

The data was collected from different datasets including Customer Data, Product Data, Store Data, and Sales Data.
The following cleaning steps were performed:
Removed duplicate values to maintain data accuracy.
Corrected formatting errors using Excel number formatting options.
Replaced incorrect values using the Replace Value feature.
Updated the data types for columns such as Date, Currency, Text, and Numbers.
Handled missing values (data imputation) in different tables.
Highlighted the cells where data was imputed for transparency.
Additional improvements:
Filled missing values in Cost and Stock columns in the product table.
Filled missing values in Quantity, Unit Price, Discount, and Total Amount in the sales table.
Added new columns such as Sales Performance, Category, Product Name, Region, Loyalty Level, and Month using the VLOOKUP function.

2. Descriptive Statistics

Descriptive statistics were created for the Total Amount Final column to understand sales distribution.
Key insights:
Average (Mean) Sales: ~1086.5
Median Sales: ~870.18
Standard Deviation: ~818.29 (indicates variation in sales values)
Minimum Sales: 18.45
Maximum Sales: 3908.24

Total Sales: 2,173,018.35
Total Transactions: 2000
This analysis helps understand the overall sales behavior and variability.

3. Pivot Table Analysis

Pivot tables were created to analyze sales from different perspectives:
Category-wise Total Revenue
Loyalty Level Revenue
Monthly Sales Trends
Region-wise Sales Performance
Payment Type Distribution
Top Selling Products
These pivot tables help quickly summarize large datasets and identify patterns.

4. Data Visualization

Charts were created in Excel to visually represent the pivot table results:
Category vs Total Revenue (Pie Chart)
Payment Type Distribution (Bar Chart)
Monthly Sales Trends (Donut Chart)
Region-wise Sales Performance (Bar Chart)
Loyalty Level Analysis (Line Chart)
Top Products Analysis (Bar Chart)
Visualization makes the sales insights easier to understand.

5. Sales Performance Dashboard

An interactive sales dashboard was created to present key insights in one place.
The dashboard includes:
Category revenue analysis
Payment method distribution
Monthly sales trends
Region-wise sales performance
Loyalty level analysis
Top selling products
It also includes filters for Region and Loyalty Level, allowing users to interactively analyze the data.

Conclusion

This project demonstrates how Excel can be used for data cleaning, analysis, and visualization. By using formulas, pivot tables, descriptive statistics, and dashboards, the dataset was transformed into meaningful business insights that help understand sales performance and customer behavior.